HSTS is a great option for forcing users to use HTTPS on your website, and if you can use HSTS preload, you can ensure that modern clients never send requests over HTTP. Nevertheless, HSTS preload can take months to enforce, and you won't always want to take that approach. In the meantime, if a browser makes an initial request over HTTP, it won't receive the HSTS header and may stay on HTTP! That's unfortunate, but you can mitigate the problem by redirecting insecure requests to HTTPS immediately.

28.4.2 Redirecting from HTTP to HTTPS with HTTPS redirection middleware

The `HstsMiddleware` should always be used in conjunction with middleware that redirects all HTTP requests to HTTPS.

TIP It's possible to apply HTTPS redirection only to specific parts of your application, such as to specific Razor Pages, but I don't recommend that, as it's too easy to open a security hole in your application.

ASP.NET Core comes with `HttpsRedirectionMiddleware`, which you can use to enforce HTTPS across your whole app. You add it to the middleware pipeline in Program.cs, and it ensures that any requests that pass through it are secure. If an HTTP request reaches the `HttpsRedirectionMiddleware`, the middleware immediately short-circuits the pipeline with a redirect to the HTTPS version of the request. The browser then repeats the request using HTTPS instead of HTTP, as shown in figure 28.8.